



EVIDENCE & ALTERNATIVES

Case 01 • ISS Greenhouse Module • Teacher-only model responses

VALIDATION BUILD

EVIDENCE TASK

3 • Investigate four evidence sources

Source	Model evidence	Interpretation
Crew	Roots grow in many directions; nutrient, water, and light adjustments did not fix them.	Supports orientation problem; weakens simple resource causes.
Sensors	Microgravity; other major readings nominal.	Identifies unusual condition and rejects several alternatives.
Plants	Dense tangles, wrapping, weak anchoring, odd stems.	Direct symptom of disrupted directional growth.
Logs	Statolith/gravitropism mechanism and replacement cues.	Explains cause and provides engineering options.

CAUSE TEST

4 • Test the competing explanations

- **Nutrient damage • weakened:** readings are nominal; tangling is not a strong match.
- **Wrong light cycle • weakened:** schedule/intensity are nominal; lighting does not best explain roots.
- **Defective seeds • weakened:** repeated environment-linked pattern.
- **Microgravity • supported:** matches all four evidence sources and the mechanism.

NOTES



Mechanism & Diagnosis



MECHANISM MODEL

5 • Build the mechanism

Earth gravity → statoliths **settle** → consistent signal → roots grow **downward**.

Microgravity → statoliths do not **settle in one direction** → unreliable signal → roots **curve or grow without consistent orientation**.



ACCURACY NOTE

Avoid requiring “random.” Roots may still respond to moisture, light, touch, chemicals, and internal growth programs.



DIAGNOSIS

6 • Diagnose and reject an alternative

Microgravity disrupted normal gravitropic orientation. Without a stable settling direction for statoliths, roots lacked a reliable “down” cue and formed tangled growth around the plant pillows.



REJECTED ALTERNATIVE

6 • Diagnose and reject an alternative

Nutrient damage is tempting because the plants are stunted and pale, but the sensors show nominal nutrient levels. It also does not explain the roots’ consistent directional/tangling problem.

NOTES



CER, Design & Exit Ticket



CLAIM-EVIDENCE-REASONING

7 • Claim-Evidence-Reasoning

CLAIM

The crop problem was caused by disrupted gravitropism in microgravity.

EVIDENCE

Kim reported roots that did not grow downward. Sensors showed microgravity while resources were normal. Plant inspection showed dense tangles and poor anchoring. Logs connected statolith-based sensing to root direction.

REASONING

Gravitropism depends on a consistent gravity direction. In microgravity, statoliths do not settle in one stable direction, so roots lose a reliable orientation signal. This explains the root pattern better than nutrient, lighting, or seed explanations, which are weakened by normal readings and the repeated environmental pattern.



ENGINEERING RESPONSE

8 • Supply a consistent orientation cue

Example design: Directional root channels inside the plant pillow. **Criterion:** most roots should extend into the medium rather than wrapping around the surface. **Constraint:** channels must remain lightweight and permit water and oxygen flow.

Directional light alone is a weaker root solution but can support shoot orientation as part of a combined design.



EXIT TICKET

9 • Exit ticket

Expected: Tangled roots should improve first because channels directly guide root direction and anchoring. Pale leaves may improve later if root structure improves resource access.



ACCEPTABLE VARIATION

Scoring guidance

- Students may use “apparent weightlessness,” “microgravity,” or “reduced reliable gravity cue.”
- Students do not need to describe auxin unless taught separately.
- A rotating chamber, moisture gradient, or combined guide can earn full credit when criteria and constraints are defensible.